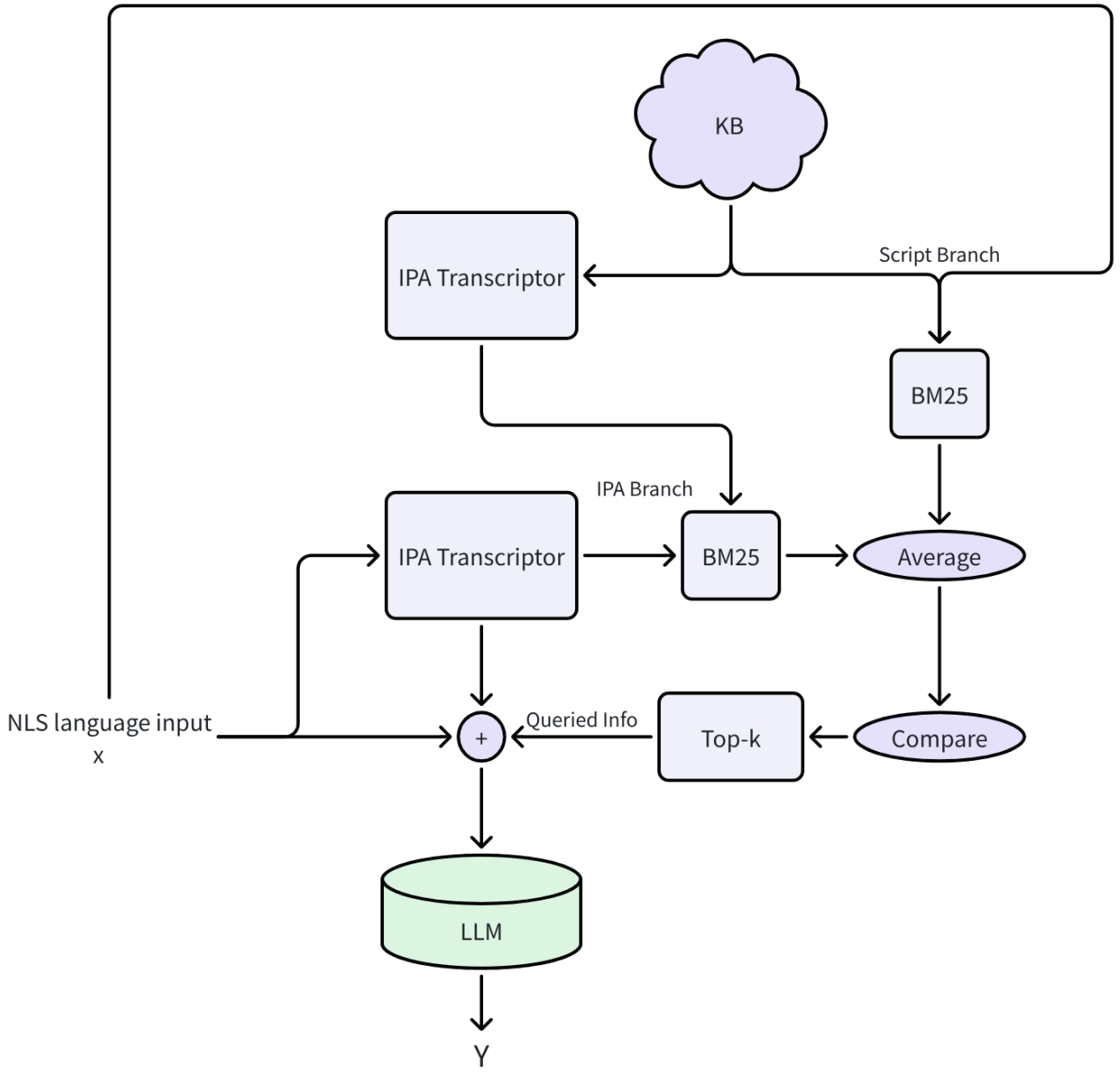
Prompting with Phonemes: Enhancing LLMs' Multilinguality for non-Latin Script Languages

> #DeepMind #Inference #Linguistics #Phonology



- Motivation
  - It is often observed that LLMs' performance will decrease when they are handling the **non-Latin script languages** such as Chinese, Japanese, Korean, Thai, and so on.
  - In the background experiment, the researchers tested non-Latin-script Languages, Latin-script Languages, and English on natural language understanding, natural language generation, translation, and QA tasks. They found that the performance of non-Latin-script languages is obviously worse than Latin-script languages and English. *(From here and on, for simplicity, Latin-Script language = LS languages, non-Latin-Script language = NLS languages.)*

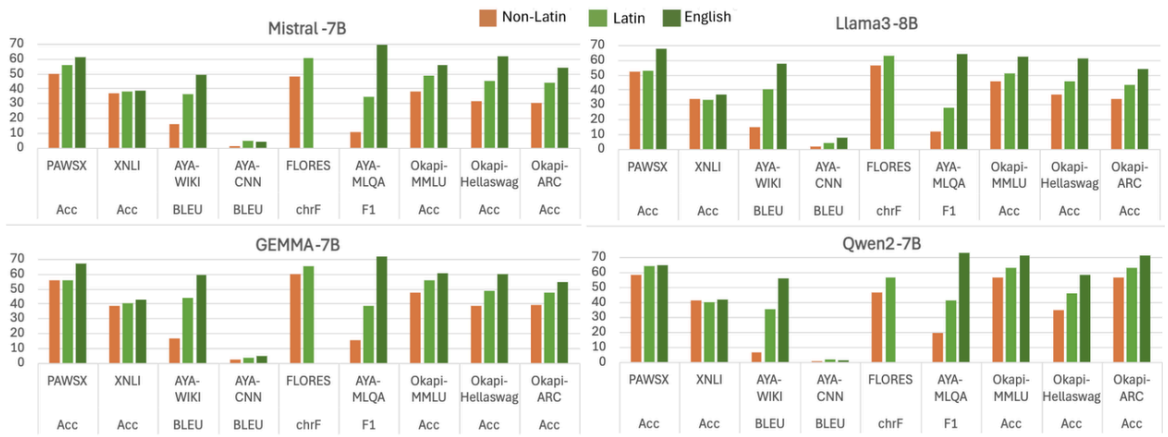


Figure 2: Performance across LLMs, grouped by languages with non-Latin scripts, Latin scripts, and English.

- Direct Prompting
  - The International Phonetic Alphabet (IPA) is ***an alphabetic system of phonetic notation*** based primarily on the **Latin script** (Wiki). In phonology, researchers use IPA as a standard representation system to describe the pronunciation of a language (no matter if it is phonogram-based or logogram-based, and no matter if it is LS or NLS).
  - Considering the experiment result listed above and the LS nature of IPA, the researchers experimented to add IPA of the input DIRECTLY to the prompt. They found that the script+IPA method of prompting **outperformed** the script-only method in NLS language tasks.

Task	Prompt Variation	Llama3-8B-Instruct		Qwen2-7B-Instruct	
		0-shot	3-shot	0-shot	3-shot
AYA-WIKI (BLEU)	Script	2.45	17.25	4.73	8.3
	Script + IPA	**2.86**	**17.76**	**8.66**	**11.73**
FLORES (chrF)	Script	38.17	**57.06**	57.96	**58.45**
	Script + IPA	**49.67**	56.87	**58.18**	58.23
AYA-MLQA (F1)	Script	17.31	20.84	15.37	27.32
	Script + IPA	**20.84**	**28.32**	**16.93**	**27.55**

- Phonemic Integration via ICL Retrieval
  - It is widely known that LLM performs better when there are examples and background information provided in the context.
  - Considering better performance using IPA, the paper proposed using IPA to retrieve background information for NLS languages, specifically, in two ways:
    - Only use IPA. Convert the input to IPA, and use BM25 to calculate the similarities, and select the piece with the top-k similarity as the context.
    - Combinational. Use both scripts and IPA to calculate the BM25 separately, calculate their average, and select the top-k ($k = 3$ in the paper) as the context.
  - It is found that the combinational method will have the best performance. It improves the effect

of both LS language tasks and NLS language tasks. On some tasks, the IPA-only method is better than script-only method, but the improvement is not very significant.

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![](<images/Paper Notes-image-76.png>)
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- It is found that the result of IPA-only and the script-only query result has only a little intersection. This indicates that the phonetic information has successfully captured cross-linguistic or deep-level feature associations that cannot be conveyed through written text.
- Plus, the combination method will have a higher possibility of keyword matching. In some experiments, it successfully matches the keywords such as "university", "address", and "management", which are crucial in answering the question.
- Based on the ablation, the mechanism with the best performance is: **IPA+Script Context & Combination Method of Query.**
- *Why don't we use embedding? Because there is no such model that embeds the IPA into a vector.*
- Romanization vs. IPA
    - *In linguistics, romanization or romanisation is the conversion of text from a different writing system to the Roman (Latin) script, or a system for doing so (Wiki).* They are usually language-specific and are not an inter-language standard, such as Pinyin (汉语拼音) and Rōmaji (Romanized Japanese).
    - This paper also explores the romanization query. It is a little better than script-only, but still less effective than the IPA-based query.
    - This is because romanization is language-specific and cannot be applied to LS languages, making it hard to do inter-language queries.

Task	Variation	Llama3-8B-Instruct		Qwen2-7B-Instruct	
		0-shot	3-shot	0-shot	3-shot
AYA-WIKI (BLEU)	Script	2.45	17.25	4.73	8.3
	Script + IPA	**2.86**	**17.76**	**8.66**	11.73
	Script + Roman	1.64	17.41	5.82	**13.30**
FLORES (chrF)	Script	38.17	57.06	57.96	**58.45**
	Script + IPA	49.67	56.87	**58.18**	58.23
	Script + Roman	**51.26**	**57.76**	57.23	58.41
AYA-MLQA (F1)	Script	17.31	20.84	15.37	27.32
	Script + IPA	20.84	28.32	**16.93**	**27.55**
	Script + Roman	**20.95**	**29.79**	13.20	27.00

- My Thoughts
    - BM25 is basically a pattern-matching algorithm, meaning that it does not include semantic information. Even though we lack an IPA-embedding model, we have script-embedding models. It

is worth trying to use **embedding similarity+BM25(IPA)** or **embedding similarity+BM25(IPA)+BM25(Scripts)** to add semantic information to the query.
    - We don't have IPA embedding model, but can we train one? As we already have multilingual datasets such as Aya Collection and FLORES, we can use tools such as Epitran to form `<Script, IPA>` pairs. Then we can use the classic transformer encoder structure as the backbone to train such an encoder. We can use **contrastive learning** with an objective such that:
        - Cross-modal alignment: Calculate the InfoNCE loss between the script embedding and phoneme embedding of the same sentence to reduce their distance in feature space.
        - Cross-lingual phoneme-semantic alignment: Utilize parallel corpora to force phoneme embeddings of semantically equivalent sentences across languages closer together.
    - Will the semantic-based IPA embeddings perform better?
- Work Cited
    - Paper
    - IPA Wikipedia
    - Romanization Wikipedia
    - Chat History
    - NotebookLM